Velocity and position graphs

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a position-time graph to velocity using slope.

SUCCESS CRITERIA - I can...

1. Read positions and elapsed times from a graph.
2. Find velocity from position change divided by time.
3. Explain the difference between speed and velocity.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “velocity” and “speed” to explain your answer. Sentence starters are optional.

velocity: rate of position change, with direction

speed: how fast something moves, without direction

slope: vertical change divided by horizontal change on a graph

01 Warm-up: bring back an earlier idea

East is positive. Start at **5 m**, go to **-3 m**, then finish at **2 m**. **Sketch the route**; **calculate distance** and displacement. Distinguish both from final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

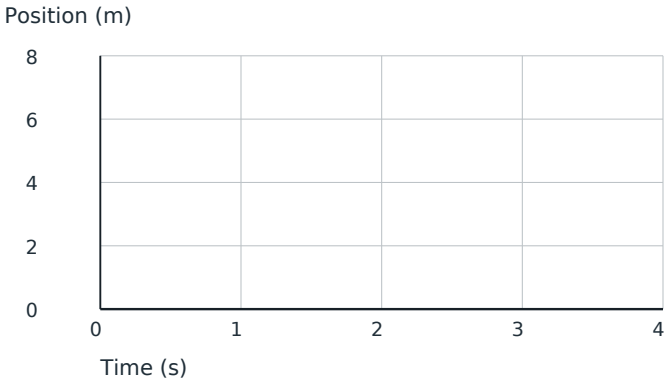
02 Worked example: explain the prediction

Reference: $v_{avg} = \Delta x/\Delta t$; average speed = distance/Δt.
Δ means change; Δt is elapsed time. Position-time slope gives velocity.

Worked reasoning: A cart moves from **1 m** to **7 m** in **3 s** at steady velocity. The position-time slope is (**7**-**1**)/**3**=**2 m/s**.

Your turn: 1. Use the blank axes to plot the cart's two measured positions and connect them.
2. **Draw a slope triangle**.
3. **Label its rise** in meters and its run in seconds.

Sentence starter: In my work, I used … to show … . I labeled it … .

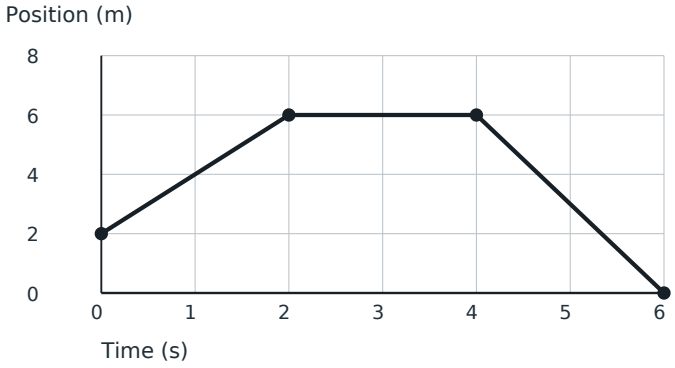


Marzano check - circle: ① ② ③

03 Interpret all three intervals

For the displayed position-time graph, calculate velocity on each segment. **Label the direction of motion** on each.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Compare two averages

For the full **6 s** trip in **3**, calculate average speed and average velocity. **Explain their difference** using the route.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

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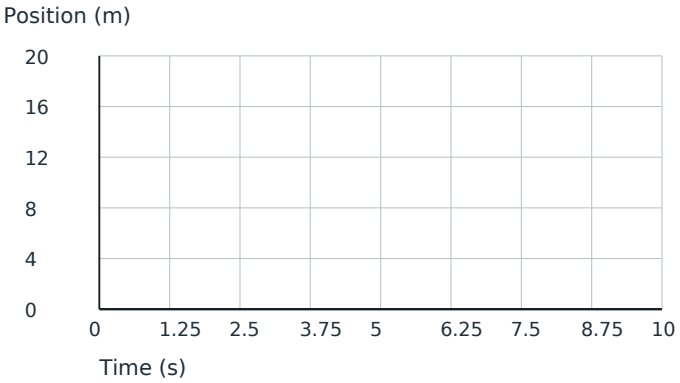
Continue down each column.

Sentence starters are optional. You may plan in any language.

05 A line that misses the origin

A robot's position-time graph passes through (**2 s**, **14 m**) and (**6 s**, **6 m**). It moves at constant velocity. Plot the line, calculate its slope, and **write x**(t). **Predict its position** at **9 s**.

Sentence starter: I predict … . I used … to make this prediction because … .

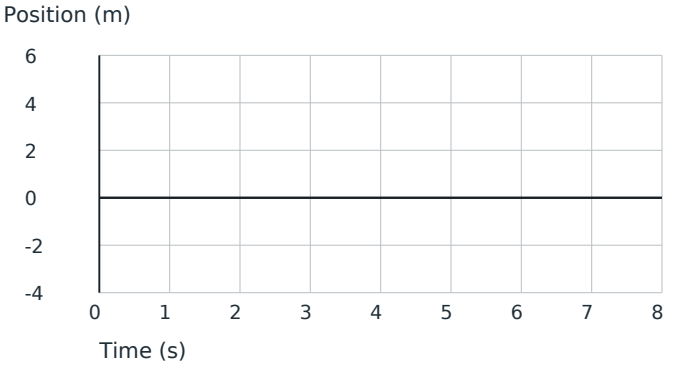


Marzano check - circle: 1 2 3

06 Construct the graph

A robot starts at **-2 m**. It moves at **+3 m/s** for **2 s**, waits for **2 s**, then moves at **-2 m/s** for **3 s**. **Draw its complete position-time graph** and **label the three intervals**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: 1 2 3

07 Separate height and slope

A student reads a point at (**5 s**, **10 m**) and reports a velocity of **2 m/s**. **What else** must be true for this calculation to be correct?

Sentence starter: I notice … . This supports … because … .

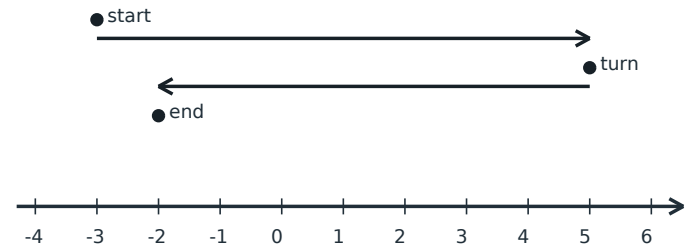
Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

East is positive. Start at **-3 m**, go to **5 m**, then finish at **-2 m**. **Sketch the route**; **calculate distance** and displacement. Distinguish both from final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.



Position (m); right is positive

Marzano check - circle: 1 2 3

09 Exit check

A straight segment runs from (**1 s**, **-2 m**) to (**5 s**, **+6 m**).
1. **Find velocity** and **state what the slope units** mean.
2. **Predict x** at **7 s** if the same motion continues.

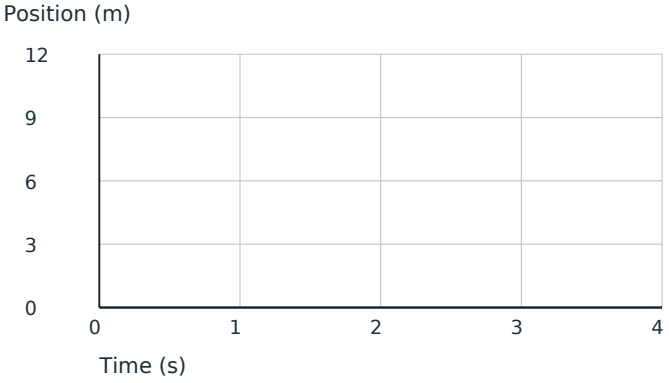
Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: 1 2 3

10 STRETCH: test the prediction

Two carts follow x_A=**1**+**2** t and x_B=**9**-t, in meters and seconds. **Sketch both graphs** and find when and where they meet.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3